



FreightX

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The Stablecoins Commerce Stack Challenge

Project Title: FreightX — Logistics & Trade Finance Orchestrator

Description: An end-to-end stablecoin-powered logistics escrow, invoice factoring, and pre-shipment financing platform built on Arc Network using Circle's USDC.

Track 2: Best SME Trade Finance & Working Capital Workflow (Invoices, Escrow, Settlement)

Circle Account Email: jessicamillervi@gmail.com

Product Feedback

1. **Products & SDKs Used:** USDC on Arc Network, Circle Developer-Controlled Wallets, Circle Gateway, USYC Treasury Sweep, StableFX, Nanopayments (ERC-8004 & ERC-8183), Circle App-Kit & Bridge-Kit (CCTP).

2. **Circle Product Feedback:**

What Worked Well: Building on Arc Testnet was smooth...

What Could Be Improved: CORS Limits on Direct IPFS Uploads... AccessControl Revert Visibility...

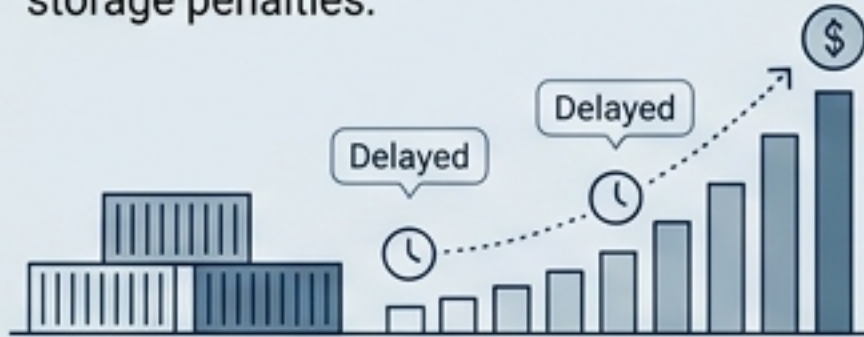
Working MVP: <https://freight-x.vercel.app/>

Video Demo Link: https://drive.google.com/file/d/1J_uZifvlevMb_NN6TY8GSMp5ToM56UYS/view?usp=sharing

Financial

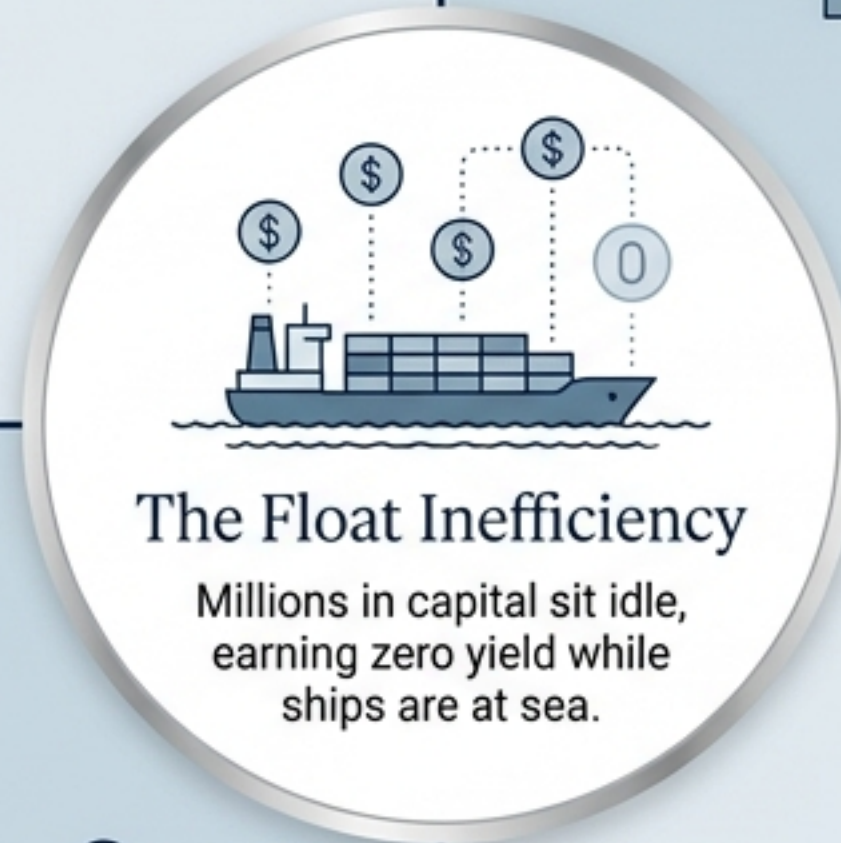
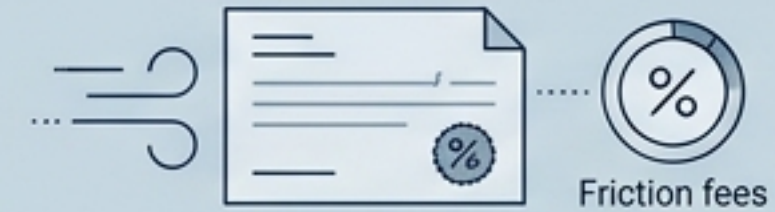
Demurrage Extortion

Manual cargo retrieval tracking triggers astronomical, delayed, and easily manipulated port storage penalties.



Working Capital Deadlock

SME suppliers lock \$30K–\$500K for 30–90 days. Sluggish bank Letters of Credit extract 1.5%–3% in pure friction fees.

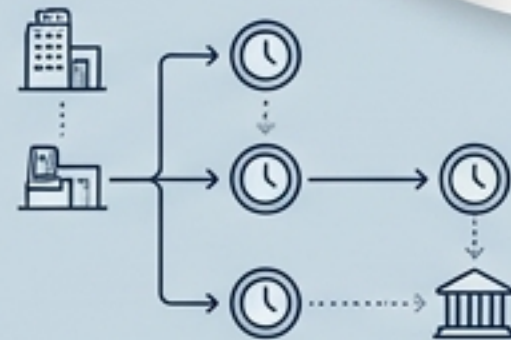


Manual

Opaque

Paper Bill-of-Lading Workflows

Reconciliation across shippers, ports, and customs takes weeks using physical, forgeable paper slips.



Cold-Chain Vulnerability

Unverified temperature conditions ruin high-value perishables, leading to months of locked-capital legal disputes.



Operational



Smart Escrows

Buyer funds locked on-chain in USDC/EURC, auto-released via code as shipping milestones are cleared.



Capital Marketplace

Pre-shipment PO financing and invoice factoring unlocks immediate operating liquidity for SMEs.



AI & IoT Telemetry

Real-time container telemetry triggers autonomous dispute resolutions and cold-chain penalties instantly.



Enterprise Passports

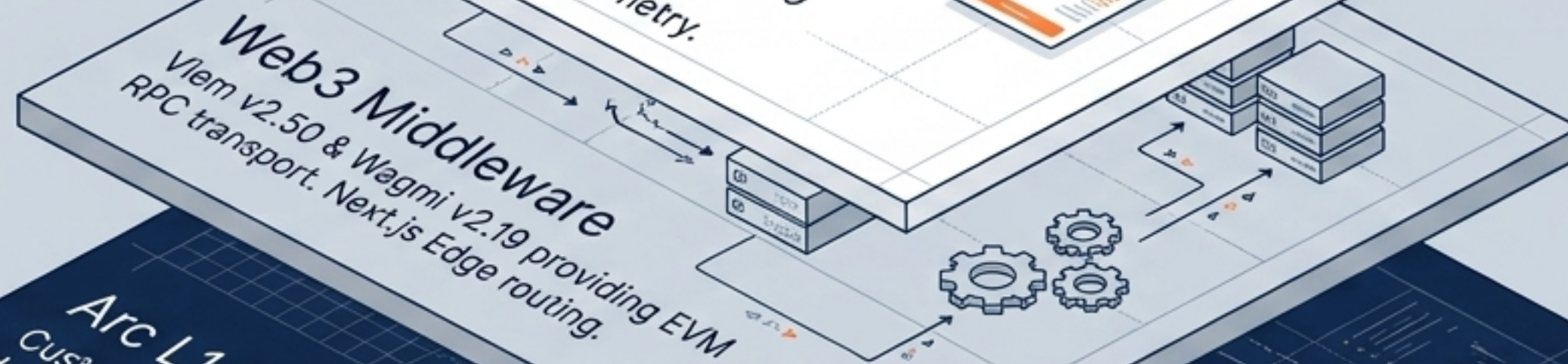
Cryptographic, verifiable reputation scoring and digital cargo twins replace paper trails.

FreightX Core (Arc L1)

Replaces legacy trade finance with programmable stablecoin escrows, executing with sub-second finality and native USDC gas.

Dimension	Legacy Trade System	FreightX Arc-Powered Solution
Asset Tracking	Paper Bill of Lading	ERC-721 Digital Twin Passport
Trade Finance	Bank Letters of Credit (5-15 days)	Programmable Smart Escrow (Instant)
Working Capital	Locked & Idle for weeks	USYC Swept (5% APY active yield mid-transit)
Dispute Resolution	Manual Arbitration	Automated IoT Telemetry & Immutable Penalties
Currency & Settlement	Fragmented, volatile FX	USDC / EURC on Arc L1 with sub-second finality

Circle Stack Integrations



USDC / EURC 
Native gas & multi-currency settlement.

USYC 
Tokenized treasury for yield sweeping.

Developer Wallets 
Powering AI Agent oracles. 

StableFX 
Integrated AED-to-USD routing.

Phase 1: Pre-Shipment PO Financing & Invoice Factoring

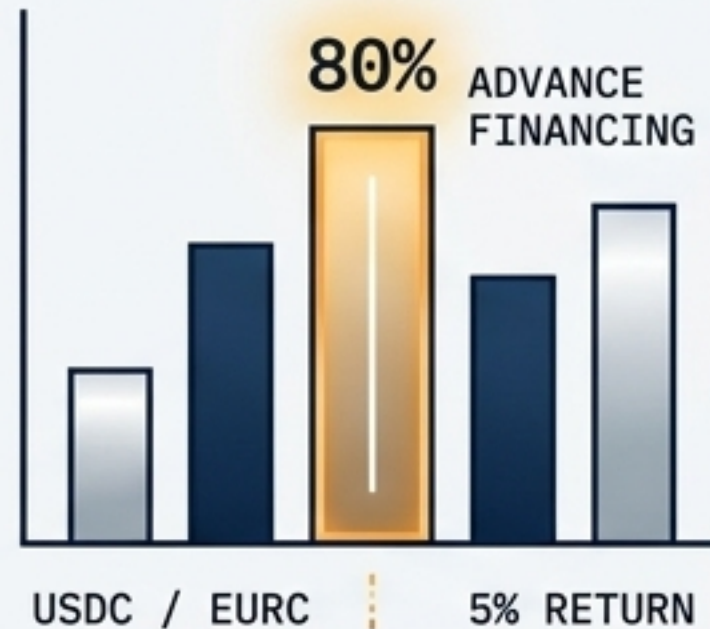
Step 1: The Request

Suppliers list Purchase Orders or pending cargo invoices on the Marketplace to unlock production liquidity.



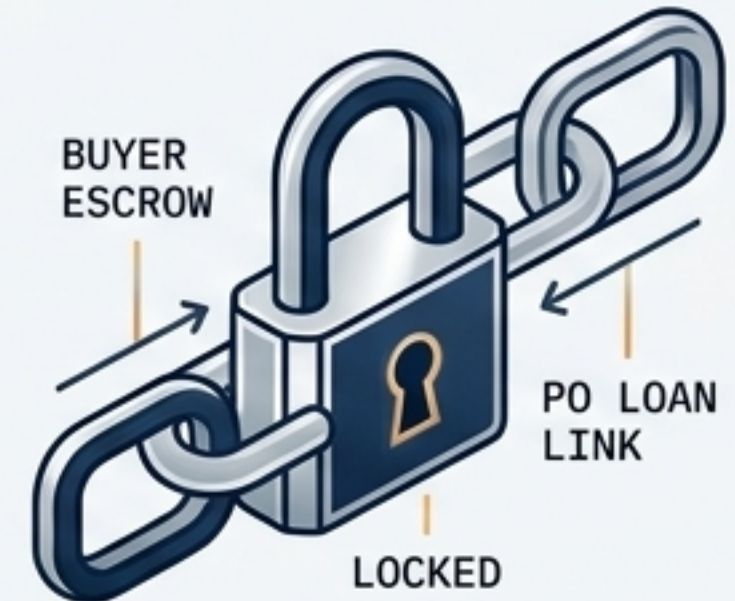
Step 2: The Funding

External investors fund POs instantly in USDC/EURC, securing up to 80% advance financing for a fixed 5% return.



Step 3: Automated Repayment

Buyer escrow deposits automatically link to the active PO loan, locking investor repayment into the final settlement.



Phase 2: Cargo Digital Twins & Smart Escrows

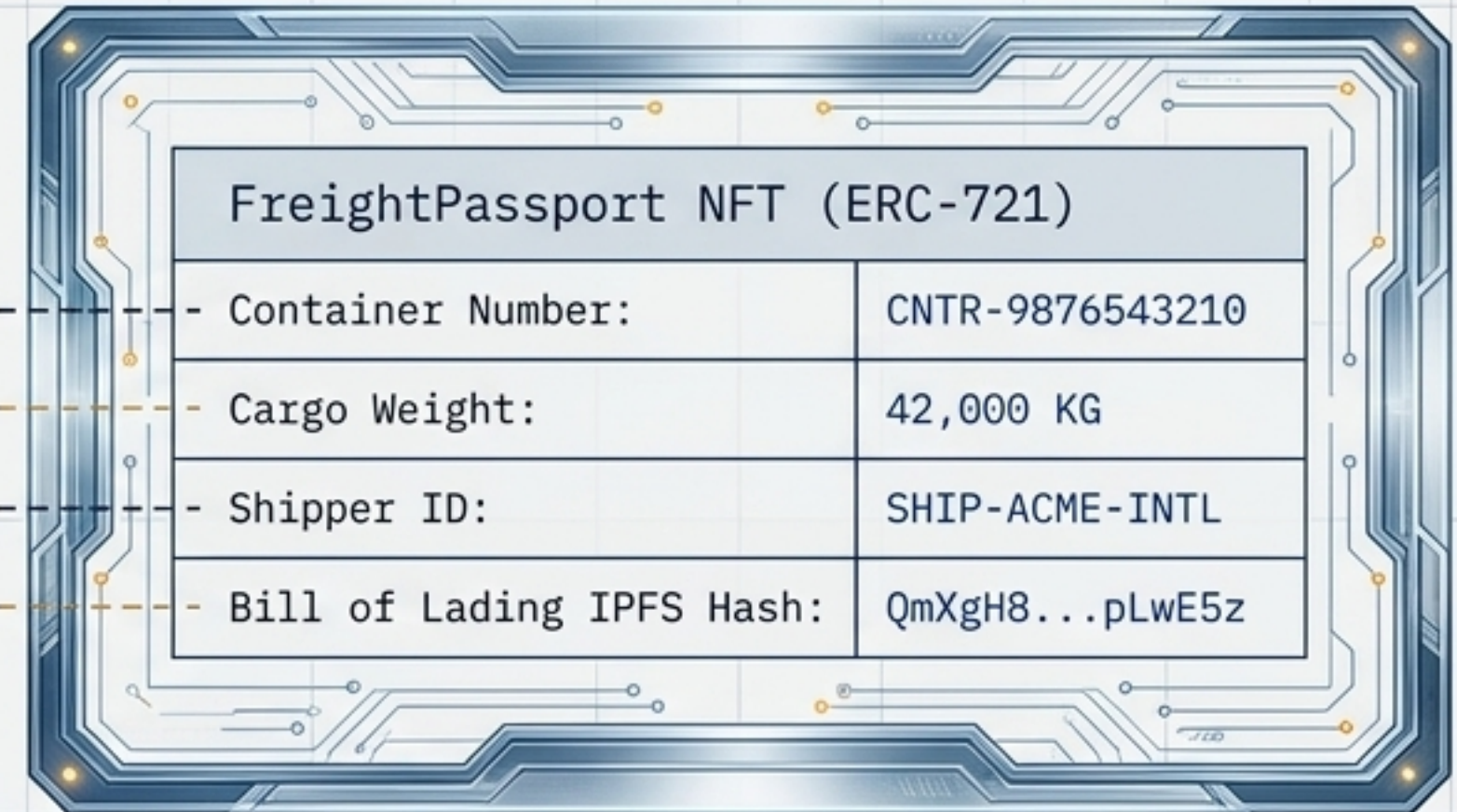
The Physical



The Escrow Lock

Buyers book the shipment and lock total cargo value plus shipping fees into the **FreightEscrow** contract.

The Digital



The Digital Twin Mint

An **ERC-721 FreightPassport NFT** is minted concurrently, acting as the immutable digital twin.

The Capital Engine: Sweeping Idle Treasuries

The Float Inefficiency

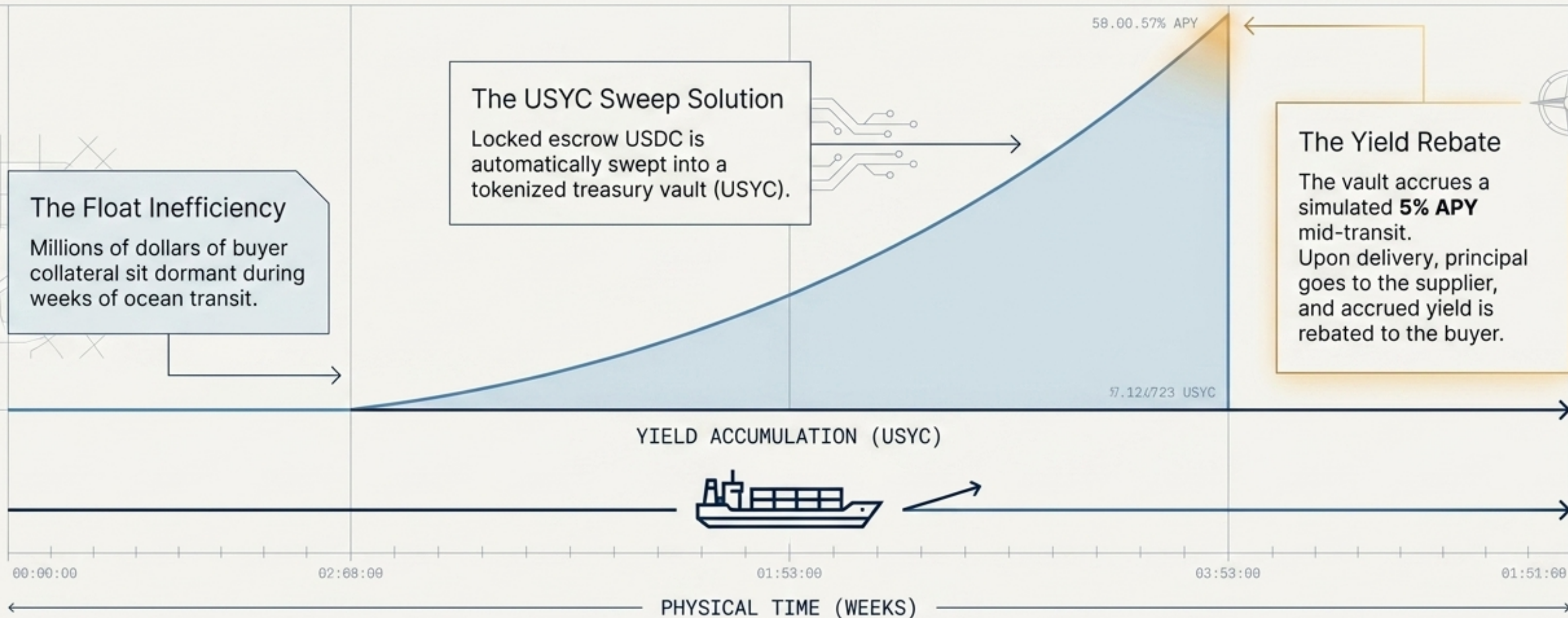
Millions of dollars of buyer collateral sit dormant during weeks of ocean transit.

The USYC Sweep Solution

Locked escrow USDC is automatically swept into a tokenized treasury vault (USYC).

The Yield Rebate

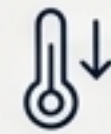
The vault accrues a simulated **5% APY** mid-transit. Upon delivery, principal goes to the supplier, and accrued yield is rebated to the buyer.

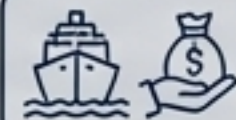


Phase 3: Autonomous IoT Telemetry & Nanopayments

Telemetry Feed

TIMESTAMP	SENSOR_ID	GPS_COORDS	TEMP (°C)	HUMIDITY (%)	STATUS
2024-10-26 14:30:15	IOT-SENS-001	01°17'N, 103°51'E	5.8	45	OK
2024-10-26 14:35:20	IOT-SENS-002	01°17'N, 103°51'E	6.1	46	OK
2024-10-26 14:40:25	IOT-SENS-001	01°17'N, 103°52'E	6.9	48	OK
2024-10-26 14:45:30	IOT-SENS-003	01°17'N, 103°52'E	8.2	50	TEMP_BREACH

 **Thermal Breach Penalties:** Sensors detect a cold-chain breach (>8.0°C). Smart contract automatically applies a predefined 5% deduction to the supplier's payout.



Mid-Transit Milestones: GPS confirms arrival at the Singapore Transshipment Hub. Contract instantly releases 30% of locked funds to the supplier.

AI Logic & Transactions



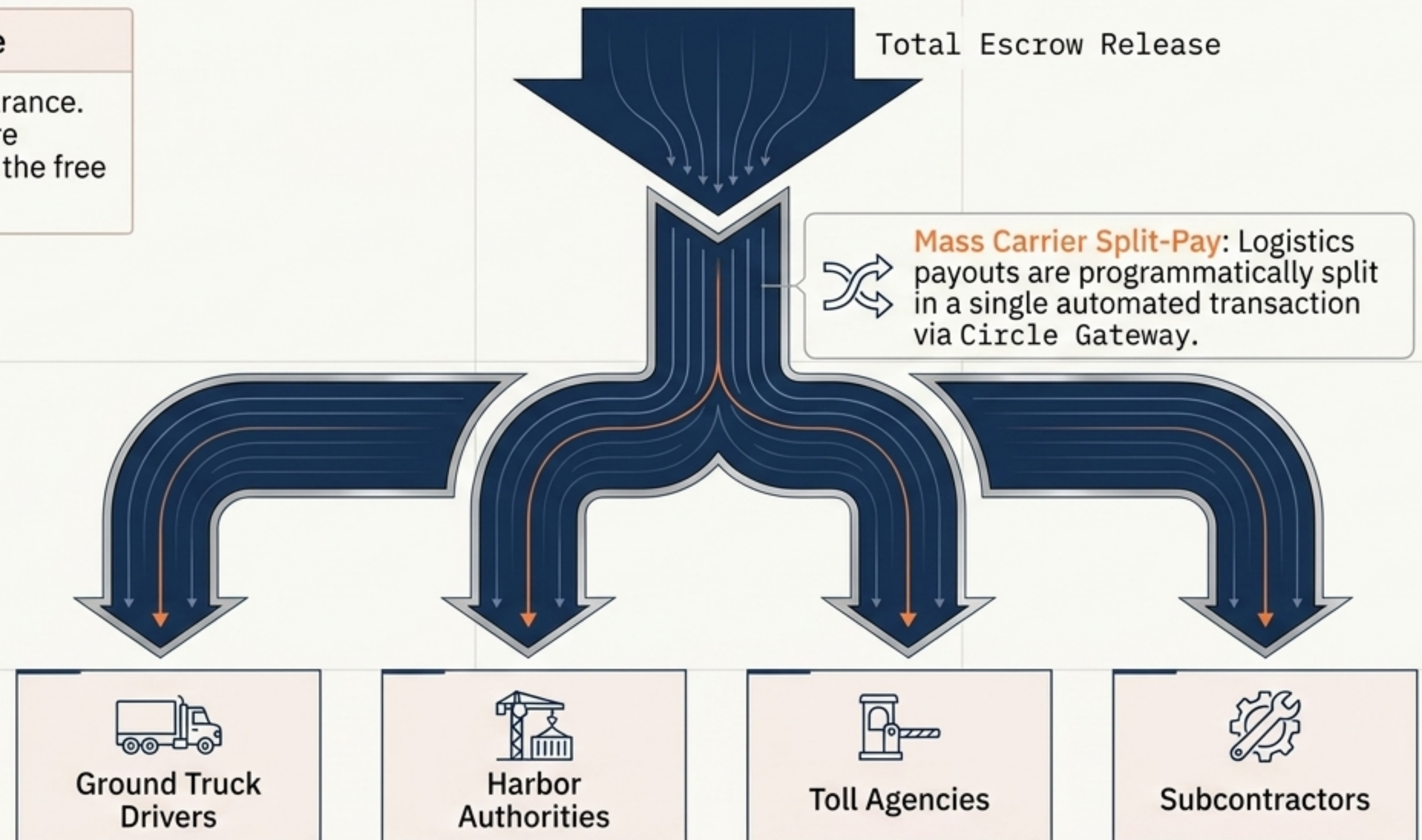
AI Logistics Coordinator: An autonomous agent executes sub-cent USDC nanopayments to unlock server-side IoT telemetry logs (HTTP 402 Paywalls).

```
1 // AI Logistics Coordinator: Autonomous Nanopayment Execution
2 POST /api/v1/telemetry/logs/access
3 Headers: {
4   "Authorization": "Bearer [AI_AGENT_TOKEN]",
5   "Content-Type": "application/json"
6 }
7 Body: {
8   "containerId": "CNTR-9876543210",
9   "logType": "FULL_TELEMETRY"
10 }
11
12 <-- RESPONSE 402 Payment Required
13 {
14   "error": "HTTP 402 Payment Required",
15   "message": "Access to full logs requires micropayment.",
16   "amount": 0.005,
17   "currency": "USDC",
18   "paymentAddress": "0x[SMART_CONTRACT_PAYWALL_ADDRESS]"
19 }
20
21 // AI Agent triggers automated payment
22 TX_HASH = sendNanopayment("0.005", "USDC", "0x[SMART_CONTRACT_PAYWALL_ADDRESS]");
23
24 --> SUCCESS 200 OK (Access Granted)
25 {
26   "status": "SUCCESS",
27   "message": "Payment verified. Access granted.",
28   "logData": { ... }
29 }
```

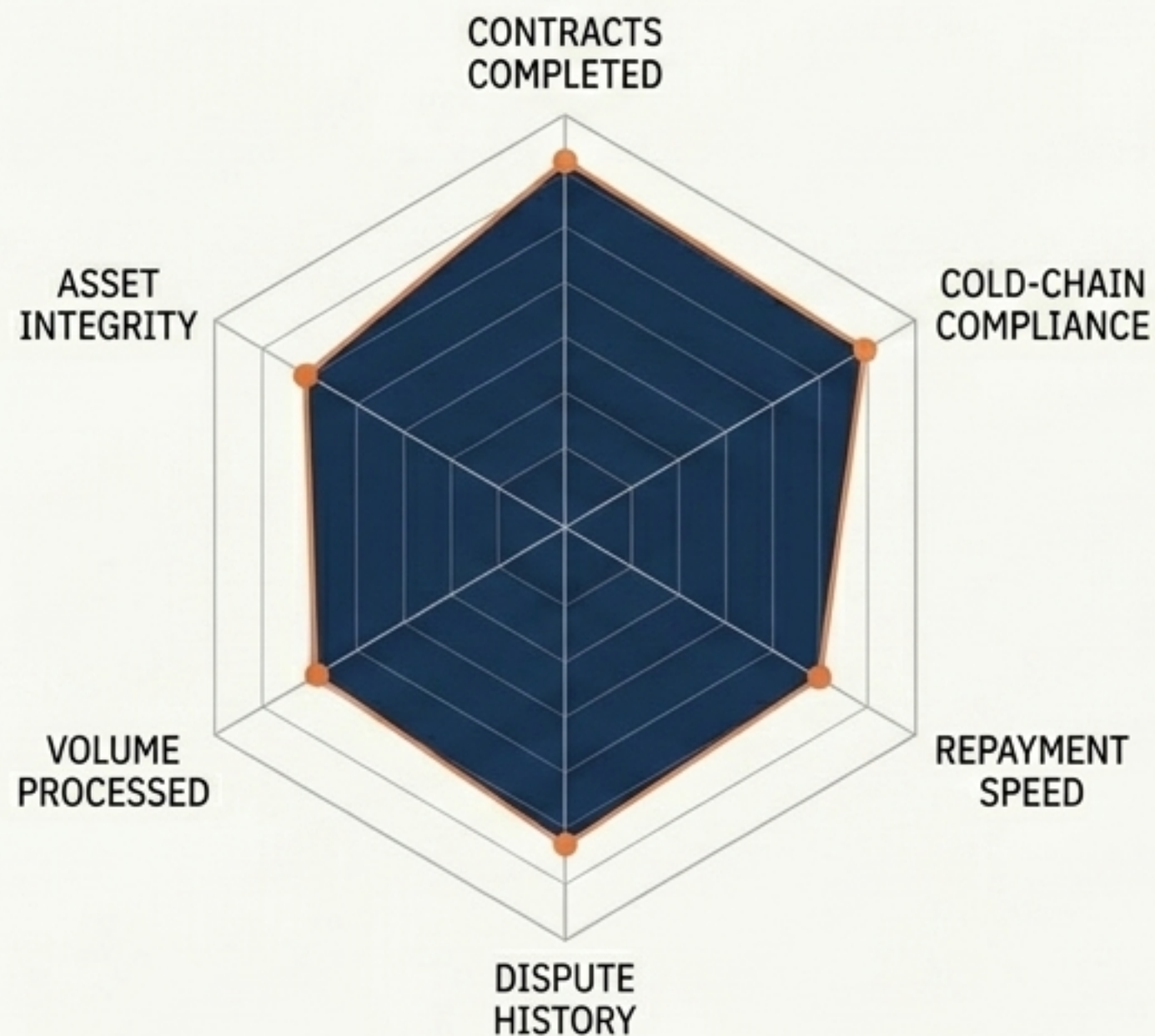

Phase 4: Demurrage Acceleration & Settlement Waterfall

Time-Accelerated Demurrage

Timers start upon Customs Clearance. Progressive storage penalties are auto-deducted if cargo exceeds the free discharge window.



Cryptographic Reputation Passports



On-Chain Analysis

Protocol analyzes metadata (completed contracts, cold-chain compliance, repayment speed) to generate dynamic enterprise grades.

```
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://www.w3.org/2018/credentials/examples/v1"
  ],
  "id": "http://example.edu/credentials/3732",
  "type": ["VerifiableCredential", "ReputationPassport"],
  "issuer": "did:example:logisticsDAO:1234",
  "issuanceDate": "2024-10-27T19:23:24Z",
  "credentialSubject": {
    "id": "did:example:carrier:5678",
    "reputationScore": "94.5",
    "coldChainCompliance": "98%",
    "completedContracts": "150+"
  },
  "proof": {
    "type": "RsaSignature2018",
    "created": "2024-10-27T19:23:24Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:example:logisticsDAO:1234#key-1",
    "jws": "eyJhbGciOiJIUzUxMiIsImImlzIjoiMjRCI6znfsc20s10lsijY8Tl19..."
  }
}
```

W3C Verifiable Credentials

1-click export of reputation as a W3C-compliant Verifiable Credential (JSON-LD format) to prove trade history off-chain.

Developer Experience: Building on Arc L1



What Worked Exceptionally Well

- **Deterministic Gas**
Arc's dollar-denominated USDC gas eliminated volatility, making transaction costs highly predictable.
- **Sub-Second Finality**
Speeds perfectly matched the immediate pacing required for physical freight processing.



Circle Product Feedback & Requests

- **Unified Faucets**
A single testnet faucet dispensing all Circle assets (USDC & EURC) would accelerate multi-currency testing.
- **StableFX Sandbox API**
Public sandbox access with mock rates would enhance real-time FX-aware escrow development.

Open Source Governance & Security Safeguards



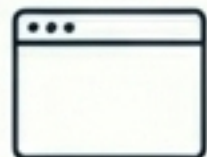
Community Standards

Governed by a strict Contributor Covenant Code of Conduct ensuring a diverse, inclusive developer environment.



Private Key Isolation

Deployer keys are strictly excluded from git tracking.



Browser Sandboxing

Interactive sandbox generates and isolates private keys purely within browser-level localStorage.



Multi-Step Oracles

Escrows require cryptographic signatures from registered IoT gateways before capital disbursements.

The Future Horizon

Waypoint 4: Multi-Sig Arbitration



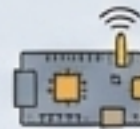
On-chain panel of independent cargo surveyors for complex physical disputes.

Waypoint 3: AI Underwriting



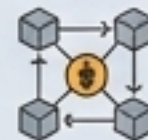
Decentralized credit scoring for optimized factoring discount premiums.

Waypoint 2: Production Telematics



Integrating physical Raspberry Pi Zero/ESP32 cellular sensor nodes.

Waypoint 1: Cross-Chain CCTP



Seamless USDC payment routing across Ethereum, Arbitrum, Solana, and Arc.

Built for the Agora Hackathon — USDC Commerce Stack Challenge

Developed with insight and energy by
Jessica Miller & Team.

GitHub: @jessicamillervi

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Explore the live FreightX sandbox on the Arc Testnet.